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Cogs 209 Mini-Project #1

Revised:

Predicting AI Tool Adoption from Socioeconomic Status:
Is Artificial Intelligence Becoming a New Axis of Privilege?

We would like to thank all three reviewers for their careful and constructive engagement with our project. Their feedback substantially strengthened the paper, and we have made several meaningful revisions in response. Following Huang and Cohen's shared concern that our original analysis did not directly test our hypothesis, we added a logistic regression with an income x education interaction term alongside the Random Forest, allowing each method to serve its predictive or inferential purpose. We applied survey weights (`WEIGHT_W152`) to all descriptive analyses as Huang recommended, with a new paragraph in the methods section clarifying where weights were and were not used. We also added AUC, F1, precision, and recall to Table 1, as both Huang and Cohen suggested. To address Tran and Huang's concern about data sparseness, we added distribution plots for age, income, and education, so readers can directly assess how many respondents fall into each segment, and updated the corresponding figure to visually flag sparse cells. We also expanded on additional research findings from Pew Research on the relationships between demographics and AI use adoption among teens, as Tran suggested. Our limitations section features an explicit acknowledgement of the mismatch between our binary use variable and an ordinal measure of depth or frequency of AI use, as noted by Tran and Coehn. We also added Tran's concern about family income meaning different things across education levels as a measurement limitation, and addressed Huang's concern about non-random missingness by adding a brief characterization of dropped cases in our description of methods We believe these changes collectively address the core methodological concerns raised across all three reviews while still maintaining the predictive modeling approach that motivated our original design.

INTRODUCTION

AI tools like ChatGPT, Gemini, and GitHub Copilot are increasingly used for writing, coding, research, and professional tasks. As these tools grow in influence, a critical question emerges: is adoption distributed equitably across society, or is it becoming yet another marker of socioeconomic privilege? This project investigates whether income and education level are significant predictors of AI chatbot adoption among U.S. adults, and which factor more strongly predicts that adoption.

Controlled experiments have found that workers given access to AI complete tasks significantly faster and at higher quality than those without (Noy & Zhang, 2023; Brynjolfsson et al., 2023). PwC's 2025 Global Workforce Survey of nearly 50,000 workers found that daily AI users are far more likely to report higher pay and productivity than infrequent users, and that workers with AI skills earn 56% more than peers in the same role without them (PwC, 2025a; 2025b). If adoption of these tools is already concentrated among higher-income and higher-educated groups, AI may compound existing socioeconomic inequality rather than democratize opportunity.

Education is theorized to matter through a different mechanism. In 2024, 20% of workers with a bachelor's degree or higher said at least some of their work was done using AI, while 13% of workers with some college education or less reported AI use. (Lin, 2025a). By 2025, AI use among college graduates increased 8 percentage points compared to a 3 percentage point increase among those with less education (Lin, 2025b). This education gradient may reflect the contexts in which AI familiarity develops, such as through college coursework, white-collar professional environments, and knowledge-work occupations that expose workers to AI tools earlier and more routinely than jobs that do not require a degree (Lin, 2025a).

Taken together, these patterns suggest that AI adoption may be reproducing the same socioeconomic structure that has characterized prior waves of digital technology. A growing share of U.S. workers say at least some of their work is done with AI, rising from 16% in 2024 to 21% in a September 2025 survey (Pew Research Center, 2025b). If that majority is disproportionately lower-income and less educated, the AI productivity premium documented by PwC will widen existing wage inequality rather than close it. Understanding who is and is not adopting AI is therefore relevant not only for researchers but for policymakers and educators seeking to address this gap and concerned with equitable technology access.

Using income and education to operationalize measures of socioeconomic privilege, we propose two hypotheses for AI adoption. First, that both income and education will be independently and significantly associated with AI chatbot adoption after controlling for age, gender, and race, and that education will emerge as the stronger predictor given that AI familiarity is largely mediated through professional and academic contexts. Second, that income and education will interact such that their combined effect on adoption exceeds what either variable predicts independently, individuals who are both high-income and highly educated will show disproportionately higher adoption rates than those who are high on only one dimension, reflecting a compounding privilege effect.

METHODS

Data

This study uses Wave 152 of the Pew Research Center's American Trends Panel (ATP), which is a nationally representative survey of 5,410 U.S. adults conducted between August 12–18, 2024 examining public attitudes toward and use of artificial intelligence tools. Data were collected online and by live telephone in English and Spanish using address-based sampling. The survey included an oversample of Hispanic men, non-Hispanic Black men, and non-Hispanic Asian adults, who were weighted back to their population proportions using the provided sample weight (`WEIGHT_W152`). The ATP weights are calibrated to match U.S. Census Bureau benchmarks for gender, race, ethnicity, education, and partisan affiliation (Pew Research Center, 2025a). In the raw unweighted sample, Hispanic respondents comprised 13.1% of observations but 8.2% after weighting, reflecting the oversample correction. The dataset is publicly available at the Pew Research Center Dataset's page following free account registration.

Of the 5,410 respondents in the full ATP Wave 152 sample, 1,612 (29.8%) were excluded, producing a cleaned sample used in this study of n = 3,798. The large majority of the exclusions, 1,375 respondents, were from survey routing. Those who reported being not at all familiar with AI chatbots on the awareness screener (`CHATAWARE_W152` = 3) were not asked the chatbot use question and therefore have no value on the primary outcome. An additional 8 refused to answer `CHATUSE_W152`. The remaining exclusions reflect refusal to answer demographic items: income (n = 260), AI use frequency (n = 33), race (n = 31), age (n = 21), gender (n = 18), and education (n = 13), with some overlap across categories.

Therefore, the sample used for this analysis is best understood as representing U.S. adults who are at least somewhat aware of AI chatbots, instead of accounting for the full adult population. Excluded respondents were substantially lower in education, lower in income, and older than those retained in the dataset. Excluded respondents averaged "some college" (M = 3.03) versus "associate degree" (M = 4.10) for the analytic sample on education, and roughly $50–60k (M = 4.24) versus $70–80k (M = 6.08) on income. Nearly half of excluded respondents were 65 or older (44.7%) compared to 19.4% of retained respondents, and excluded respondents were more than twice as likely to fall in the lowest income bracket (25.1% vs. 11.5%). These differences reflect the fact that AI chatbot awareness itself follows a socioeconomic gradient and the respondents excluded by survey routing are precisely those least likely to have adopted AI tools. The findings should be interpreted within that scope..

The primary outcome is `CHATUSE_W152`, recoded as a binary indicator: 1 = has ever used an AI chatbot (ChatGPT, Gemini, or Copilot); 0 = has not. The primary predictors are education (`educ_6cat`: 1 = less than HS through 6 = postgraduate) and family income (`inc_9cat`: 1 = under $30k through 9 = $130k+). They are already coded as ordered categories in the original dataset so no manual ordinal recoding was performed. Both are treated as ordinal integers in all models, as their rank ordering carries meaningful directional information. Demographic controls include age category (`age_cat`: 1 = 18–29 through 4 = 65+), gender (reference: men), and race (reference: White non-Hispanic). Gender and race are dummy-coded using drop-first encoding to avoid perfect multicollinearity. For gender, men are the omitted reference category and for race, White non-Hispanic is the omitted reference category. All descriptive statistics and inferential models were estimated using the ATP survey weight (`WEIGHT_W152`) to produce population-representative estimates. Survey weights were not applied to cross-validated predictive accuracy comparisons, as model comparison metrics are not affected by weighting.

Computational and Statistical Methods

All analyses were conducted in Python 3.11 using scikit-learn, statsmodels, pandas, scipy, seaborn, and matplotlib. Each analytic method was selected to answer a distinct research question.

> Distribution checks: Prior to modeling, the distribution of each predictor was examined using a kernel density plot for age and bar charts for education and income. Raw category counts were inspected to identify sparse cells, defined as n < 30, that could produce unstable predicted probabilities in the adoption heatmap.
>
> Logistic regression: Three weighted logistic regression models were estimated using statsmodels GLM with a binomial family and variance weights to provide full inference output. Model 1 included education and income only. Model 2 added age, gender, and race as demographic controls. Model 3 added an income x education interaction term to directly test Hypothesis 2. An odds ratio greater than 1.0 with a confidence interval not crossing 1.0 indicates a positive, statistically significant association. Model fit was compared using AIC, with lower values indicating better fit, and McFadden's pseudo-R^2 is reported for Models 1 and 2.
>
> Random Forest classifier: A Random Forest classifier (Breiman, 2001) was used to assess overall predictive accuracy rather than to quantify predictor-specific effects. It accommodates ordinal predictors without dummy coding and can capture non-linear relationships, such as the hypothesized income × education effect. Models 4 and 5 used 100 trees, no maximum depth constraint, and a fixed random seed (42) for reproducibility. Prior to cross-validation, ordinal and dummy-encoded versions of the

socioeconomic predictors were compared on a held-out test set to confirm that encoding choice did not affect the classifier performance.

Model comparison: The four models that were evaluated using 5-fold cross-validation are logistic regression with socioeconomic predictors only (Model 1), logistic regression with the full demographic model (Model 2), Random Forest with socioeconomic predictors only (Model 4), and Random Forest with the full demographic model (Model 5). Mean accuracy and standard deviation across folds are reported alongside AUC, F1, precision, and recall for each. AUC is reported as a secondary metric measuring how well each model separates adopters from non-adopters. A naive majority-class baseline was computed from the marginal class distribution of `chatuse_binary` to provide a performance reference by predicting the most common class for every observation.

Permutation importance: Permutation importance was used on a held-out test to measure how much predictive accuracy declines when each feature's values are randomly reassigned across respondents. A larger accuracy decline indicates the model relied more heavily on that feature. For nominal variables represented by multiple dummy indicators, all indicators for a given variable were shuffled simultaneously to measure the total predictive contribution of race and gender as categories rather than the marginal contribution of individual dummy indicators.

Secondary outcome: Ordinal logistic regression models were estimated using statsmodels OrderedModel with two specifications, SES predictors only and the full demographic model, mirroring the adoption analysis structure. Weighted Spearman rank correlations are reported as a nonparametric descriptive check.

Adoption probability heatmap: To illustrate the income-by-education adoption gradient, the socioeconomic Random Forest model was fit on the full analytic sample and used to generate predicted adoption probabilities across a complete grid of all 54 income × education combinations via `predict_proba`, displayed as an annotated heatmap. A companion cell-count heatmap is displayed alongside the adoption heatmap, with cells containing fewer than 30 observations flagged with an asterisk to show data sparsity.

Survey Weights: Survey weights were applied to all descriptive statistics and inferential models to produce population estimates representative of the U.S. adult population. Weights were not applied to cross-validated predictive models, as model comparisons are unaffected by weighting. Thus, predictive model outputs reflect the analytic sample and should not be interpreted as population-level estimates.

RESULTS

Predictor Distributions

Inspection of predictor distributions prior to modeling revealed meaningful skew in all three variables (Figure 1). The age distribution was dominated by the 30–49 category (n = 1,493), with the youngest group (18–29) underrepresented (n = 633). The less-than-high-school education cell was sparse (n = 109), contributing to the heatmap instability discussed in Section 3e. Income was heavily right-skewed, with the $130k+ category (n = 1,514) nearly six times larger than the $30–40k category (n = 252). This could possibly reflect that lower-income respondents are more likely to report household income shared across intergenerational living arrangements, while higher-income respondents more often report individual earnings. These distributional features are noted as limitations for the heatmap interpretation and for population-level generalization.

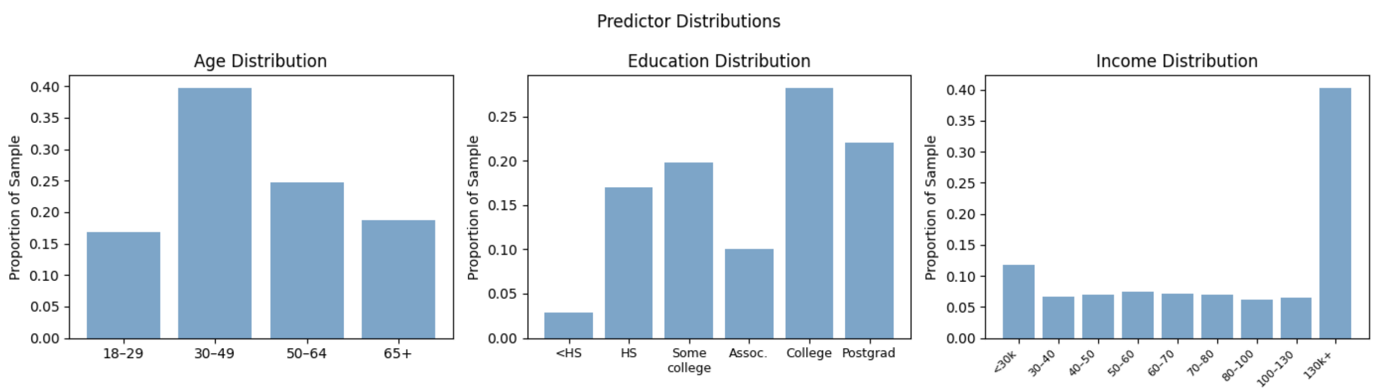

Figure 1. Unweighted distribution of age, education, and income categories in the analytic sample (n = 3,758), expressed as proportion of total respondents.

Weighted Adoption Rates

Before adjusting for other variables, weighted adoption rates show clear gradients across all three predictors (Figure 2). Adoption increased steadily with education, from 27.9% among those with less than a high school education to 59.8% among postgraduates. The income gradient was less consistent, with adoption rates relatively flat between $40,000–$100,000 before rising sharply above $100,000 (54.7% and 57.2% in the top two brackets). The starkest gradient was by age where adoption ranged from 63.3% among 18–29 year-olds to just 23.8% among those 65 and older, a 40-percentage-point gap larger than any education or income difference.

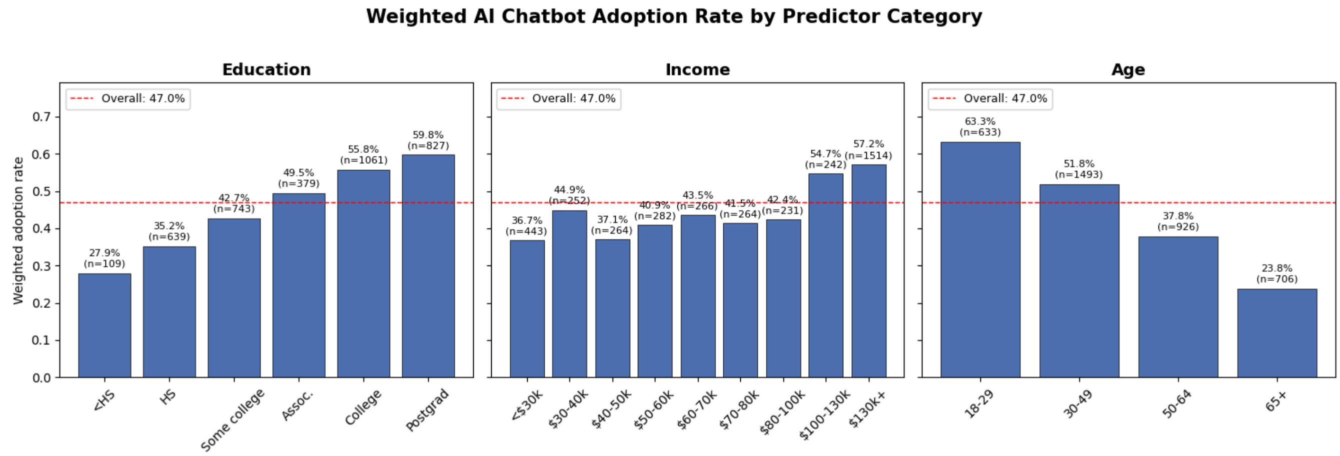

Figure 2. Weighted AI chatbot adoption rate by education, income, and age category with sample sizes (n) shown above each bar. The dashed red line indicates the overall weighted adoption rate (47.0%)

Logistic Regression

Table 1 presents the results of the three logistic regressions models predicting AI chatbot adoption, where Model 1 includes only education and income, Model 2 adds age, gender, and race as demographic controls, and Model 3 adds an interaction term between education and income.

Education was independently and significantly associated with AI chatbot adoption across all three models, while income's association was significant in Models 1 and 2 but weakened to non-significance once the interaction term was introduced in Model 3. In Model 2, the full weighted demographic model, education showed the strongest association (OR = 1.296, 95% CI [1.231, 1.365], $p < .001$), indicating that each one-level increase on the six-level education scale is associated with a 30% increase in the odds of adoption, controlling for income, age, gender, and race. Income also showed a significant and positive association (OR = 1.070, 95% CI [1.042, 1.099], $p < .001$), indicating that each one-level increase on the nine-level income scale is associated with a 7% increase in the odds of adoption, controlling for all other variables. The larger odds ratio for education relative to income, along with education's continued significance in Model 3 where income was not, supports Hypothesis 1 that education is the stronger independent predictor of AI chatbot adoption.

Among the demographic controls in Model 2, age showed the largest effect of any predictor in the model (OR = 0.528, 95% CI [0.489, 0.571], $p < .001$), meaning each one category increase in age is associated with a 47% reduction in the odds of adoption, controlling for all other variables. Women had significantly lower odds of adoption than men (OR = 0.709, $p < .001$). Among the race categories, Hispanic respondents showed significantly higher odds of adoption than non-Hispanic White respondents (OR = 1.832, 95% CI [1.396, 2.403], $p < .001$), which was the largest race effect observed in the model and no other race category reached statistical significance.

Model 3 directly tested Hypothesis 2, that the combined effect of high income and high education on adoption would exceed what either predictor contributes independently, by adding an income x education interaction term. A significant interaction term greater than 1 would have indicated that individuals high in both income and education have disproportionately higher odds of adoption than would be expected from summing their separate effects, consistent with a compounding privilege effect. Instead, the interaction was not statistically significant (OR = 1.004, 95% CI [0.988, 1.021], $p = .616$), indicating that the effects of income and education on adoption are additive rather than compounding. Income's coefficient also lost statistical significance in Model 3, likely reflecting collinearity introduced by the interaction term rather than a true change in income's relationship with adoption. Moreover, model fit did not improve with the addition of the interaction term, AIC increased slightly from 4440.0 in Model 2 to 4441.8 in Model 3, and McFadden's pseudo-R^2 remained unchanged at 0.105. Altogether, these results indicate that Hypothesis 2 is not fully supported and there is no evidence that high education and high income jointly produce adoption rates exceeding what would be expected from their independent effects alone.

Table 1. *Logistic Regression Results: Odds Ratios and 95% Confidence Intervals Across Models*

Predictor	Model 1 (SES only)	Model 2 (+ demographics)	Model 3 (+ interaction)
Intercept	0.28*** (0.23–0.34)	1.07 (0.82–1.39)	1.16 (0.76–1.77)
Education	**1.24*** (1.19–1.31)**	**1.30*** (1.23–1.36)**	**1.26*** (1.13–1.41)**
Income	1.06*** (1.03–1.08)	1.07*** (1.04–1.10)	1.05 (0.99–1.13)
Age	—	0.53*** (0.49–0.57)	0.53*** (0.49–0.57)
Gender (woman)	—	0.71*** (0.61–0.82)	0.71*** (0.61–0.82)
Gender (other)	—	1.18 (0.55–2.53)	1.17 (0.55–2.51)
Race (Black)	—	1.08 (0.86–1.37)	1.08 (0.85–1.37)
Race (Hispanic)	—	**1.83*** (1.40–2.40)**	**1.83*** (1.39–2.40)**
Race (other)	—	0.96 (0.70–1.32)	0.96 (0.70–1.32)
Race (Asian)	—	0.78 (0.54–1.12)	0.78 (0.54–1.11)
Income × Education	—	—	1.00 (0.99–1.02)
AIC	**4781.2**	**4440.0**	**4441.8**
McFadden's pseudo-R^2	0.033	0.105	0.105
n	3758	3758	3758

Note. Cell values are odds ratios with 95% confidence intervals in parentheses (e.g., 1.30 [1.23, 1.36] indicates OR = 1.30). Reference categories: Education (less than high school); Income (under $30,000); Age (18–29); Gender (man); Race/ethnicity (White, non-Hispanic). Model 1: education and income only; Model 2: adds age, gender, and race; Model 3: adds an income × education interaction term. AIC = Akaike Information Criterion (lower values indicate better fit). *p < .05, **p < .01, ***p < .001.

Model Accuracy

The naive majority-class baseline was 0.501, reflecting a near-perfectly balanced outcome variable. All four predictive models exceeded this baseline, with the logistic regression full demographic model performing best overall (accuracy = 0.661, AUC = 0.718), outperforming the Random Forest full model (accuracy = 0.622, AUC = 0.654). As shown in Figure 2, the error bars for the logistic regression full model do not overlap with those of any other model, suggesting its accuracy advantage is consistent across cross-validation folds. The logistic regression full model achieved an accuracy 16 percentage points better than baseline (0.661 vs. 0.501), or 31.9% relative improvement, meaning it correctly classified roughly one third more respondents than chance alone would predict. Table 2 reports AUC, F1, precision, and recall for all four models. Overall, the consistency of these metrics with accuracy suggests that the near-balanced class distribution did not cause any model to systematically favor one class over the other.

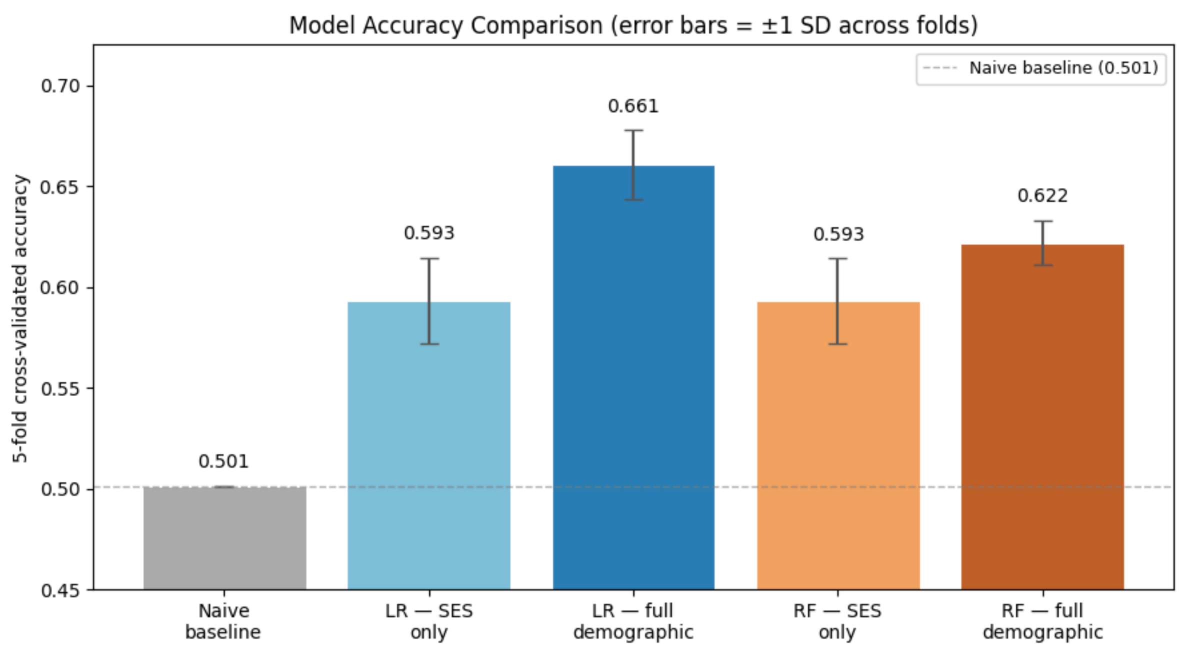

Figure 3. Five-fold cross-validated accuracy for the naive baseline and four predictive models (logistic regression and Random Forest, each with socioeconomic-only and full demographic predictors). Error bars represent ±1 standard deviation across the five folds. The dashed line marks the naive majority-class baseline (0.501). LR = logistic regression; RF = Random Forest;

SES = socioeconomic status (education and income only).

Table 2. *Five-Fold Cross-Validated Performance Metrics for All Predictive Models*

Model	Accuracy	AUC	F1	Precision	Recall
Naive baseline	**0.501**	—	—	—	—
LR — SES only	0.593 ± 0.021	0.625 ± 0.027	0.599 ± 0.025	0.593 ± 0.026	0.608 ± 0.054
LR — full demographic	**0.661 ± 0.017**	0.718 ± 0.029	0.664 ± 0.033	0.658 ± 0.017	0.676 ± 0.075
RF — SES only	0.593 ± 0.021	0.612 ± 0.030	0.617 ± 0.023	0.584 ± 0.020	0.654 ± 0.039
RF — full demographic	0.622 ± 0.011	0.654 ± 0.014	0.625 ± 0.023	0.621 ± 0.013	0.631 ± 0.052

Note. All values are reported as mean ± standard deviation across the five cross-validation folds. AUC, F1, precision, and recall are not reported for the naive baseline, as these metrics are not meaningful for a classifier that always predicts the majority class. LR = logistic regression; RF = Random Forest; SES = socioeconomic status (education and income only).

Permutation Importance

Permutation importance on a held-out test set (Figure 3) confirmed that age was the strongest predictor overall, with a mean accuracy decrease of 0.071, followed by education with 0.029, and income with 0.018. When age values were randomly scrambled across respondents, the model's ability to correctly predict AI adoption dropped by 7.1 percentage points, which is the largest accuracy loss of any predictor. Scrambling education caused a 2.9 percentage point drop, and scrambling income caused a 1.8 percentage point drop. The confidence intervals for all three variables remained above zero, supporting that each makes a meaningful contribution to the model's predictions. However, when race and gender indicators were scrambled as complete variable groups, accuracy changes were small and their confidence intervals include zero. Race showed a mean change near zero (-0.001) and gender showed a negative mean change (-0.006), indicating these variables add little predictive information to the Random Forest beyond what is already captured by age, education, and income. These results are directionally consistent with the logistic regression findings that age is the strongest demographic predictor and education contributes more than income among the socioeconomic variables.

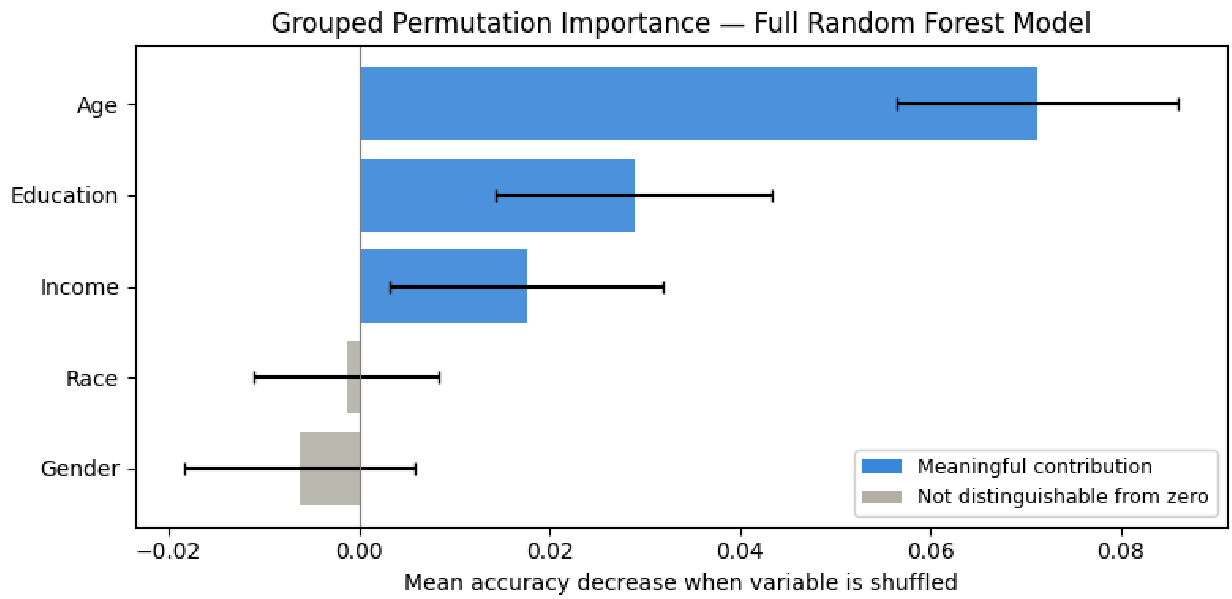

Figure 3. Grouped permutation importance for the full Random Forest model. Bars show the mean accuracy decrease when each variable is randomly shuffled in the held-out test set (±1 SD across repetitions). Race and gender were shuffled as complete groups across their dummy-coded categories. Blue bars indicate a statistically detectable contribution (CI excludes zero); gray bars indicate none detected (CI includes zero).

Adoption Probability Heatmap

Predicted adoption probabilities across the income × education grid broadly increased with both higher education and higher income, consistent with the main effects identified in the logistic regression. Among reliable cells where $n \geq 30$, postgraduate respondents showed elevated adoption probabilities across most income brackets, including two cells exceeding 0.60 ($80-100k and $130k+). The less-than-high-school row showed unreliable predictions, such as a value of 0.00 at $60-70k, due to the sparse training data rather than an underlying pattern. The row totals $n = 109$ respondents, with individual cells ranging from $n = 2$ to $n = 38$, as confirmed by the cell-count heatmap (Figure 4, top). The overall gradient across reliable cells reflects independent main effects rather than a true compounding interaction, consistent with the non-significant interaction term in Model 3.

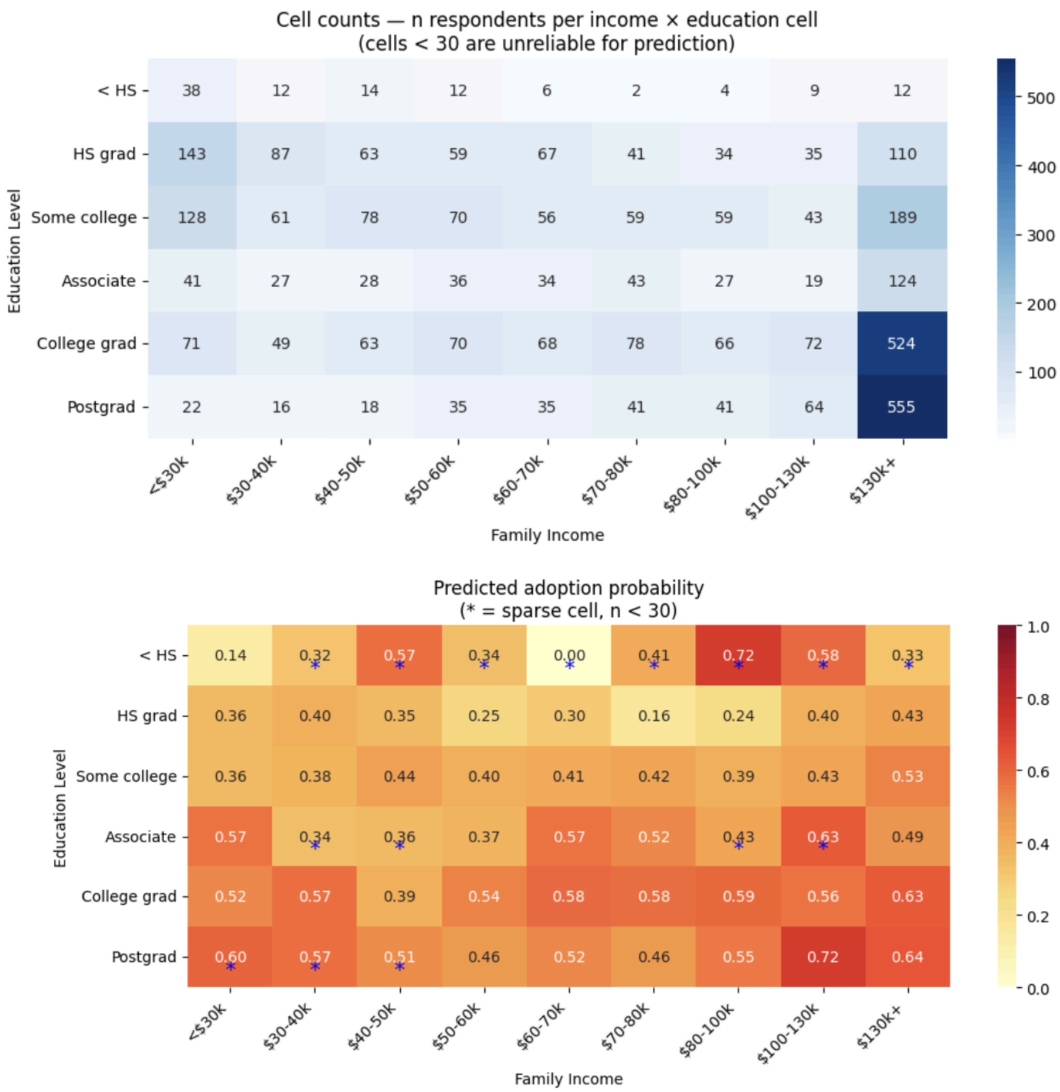

Figure 4. Predicted adoption probability and underlying cell counts across the income × education grid. Top: number of respondents (n) in each income × education combination. Bottom: predicted probability of AI chatbot adoption from the full Random Forest model for each combination, with cells flagged (*) where n < 30 and predictions are unreliable due to sparse training data.

Frequency of Use — Spearman Correlations

Both Spearman correlations were statistically significant ($p < .001$). Education showed a weak negative association with `useai` ($\rho = -0.152$) and income showed a similarly weak negative association ($\rho = -0.130$). Because `useai` is reverse-coded, with lower scores indicating more frequent use, these negative correlations indicate that higher education and higher income are both associated with more frequent AI use. The p-values confirm that these directions are reliable at n = 3,758 and are not sampling artifacts. Education accounts for approximately 2.3% of variance in use frequency and income for approximately 1.7%, indicating that most variation in how often people use AI is driven by factors not captured in this dataset.

Spearman Correlations with AI Use Frequency (`useai`)

Predictor	ρ	p
Education (`educ_6cat`)	−0.152	< .001
Income (`inc_9cat`)	−0.130	< .001

Frequency of Use — Ordinal Logistic Regression
Among AI adopters (n = 1,883), the Brant test approximation confirmed that the proportional-odds assumption held for the primary SES predictors (education SD across thresholds = 0.016; income SD = 0.016). Several demographic controls, particularly sparse race and gender categories, showed threshold instability attributable to small cell sizes rather than systematic violations; results for those predictors are interpreted with caution.

The OLR confirmed that higher education (OR = 1.092, 95% CI [1.067, 1.118], $p < .001$) and higher income (OR = 1.036, 95% CI [1.024, 1.049], $p < .001$) each independently predicted more frequent AI use after demographic adjustment. Older age predicted less frequent use (OR = 0.938, $p < .001$). Gender was non-significant, as were most race categories with two exceptions. Respondents identifying with another race/ethnicity category showed higher odds of frequent use (OR = 1.240, $p = .004$), while Asian respondents showed lower odds of frequent use relative to White non-Hispanic respondents (OR = 0.754, $p = .003$). The model explained very little variance overall (pseudo-R^2 = 0.005), indicating that SES and demographics account for minimal variation in use frequency among adopters. These results suggest that socioeconomic status plays its largest role in determining whether someone adopts AI at all, with a smaller, though still measurable, role in how often they use it once adopted.

Table 3. *Ordinal Logistic Regression: Frequency of AI Use Among Adopters*

Model	OR (95% CI)	p-value
Education	1.092 (1.067–1.118)***	<.001
Income	1.036 (1.024–1.049)***	<.001
Age	0.938 (0.905–0.972)***	<.001
Gender (woman)	1.014 (0.949–1.082)	0.687
Gender (other)	0.897 (0.682–1.181)	0.440
Race (Black)	1.053 (0.944–1.173)	0.353

Race (Hispanic)	1.036 (0.935–1.146)	0.502
Race (other)	1.240 (1.073–1.432)**	0.004
Race (Asian)	0.754 (0.627–0.906)**	0.003

Note. Outcome is frequency of AI use (useai_r), recoded so higher values indicate more frequent use (5 = almost constantly, 1 = less often), among AI adopters only (n = 1,883). OR > 1 indicates higher predictor values are associated with more frequent use; OR < 1 indicates the opposite. Reference categories: Gender = man; Race/ethnicity = White, non-Hispanic. McFadden's pseudo-R^2 = 0.005. *$p < .05$, **$p < .01$, ***$p < .001$.

DISCUSSION

Findings

Both income and education were positively associated with AI chatbot adoption and more frequent use, supporting the premise that AI adoption follows a socioeconomic gradient. However, the socioeconomic gradient appears to have a role at two stages. First through awareness, where lower-SES adults are disproportionately excluded from the analytic sample entirely, and second through adoption decisions among those who are aware. The best-performing model reached 66.1% accuracy against a 50.1% baseline, indicating meaningful predictive signal, though substantial variation remains unexplained.

Hypothesis Findings

The first hypothesis received full support. Both socioeconomic predictors were independently and significantly associated with adoption after controlling for demographics, and education emerged as the stronger predictor, consistent with the original hypothesis that AI familiarity is shaped by academic and professional contexts. The logistic regression confirmed this ordering, with education showing a larger effect on the odds of adoption than income.

The second hypothesis, that high income and high education would interact to compound adoption, was not supported. The income × education interaction term was not statistically significant, and adding it did not improve model fit. The effects of income and education on adoption are approximately additive and each contributes independently, but having both does not produce an extra advantage beyond what each contributes on its own. The visual gradient in the heatmap reflects those independent contributions rather than a true interaction effect.

Demographic Findings

Age was the strongest predictor in both the logistic regression (OR = 0.528 per age category) and permutation importance (0.071), substantially exceeding both socioeconomic predictors. The weighted descriptive statistics indicate that adoption was 63.3% among 18–29 year olds and only

23.8% among those 65 and older, demonstrating a 40-percentage-point gap larger than any education or income difference. Generational familiarity with digital tools is at least as consequential as socioeconomic status in predicting AI adoption.

An additional finding was that Hispanic respondents showed significantly higher adoption than White non-Hispanic respondents ($OR = 1.832$, $p < .001$) after controlling for education, income, and age. This was the largest race effect in the model. It may reflect cultural factors, differential occupational exposure, or the younger age profile of Hispanic adults in this sample, and warrants further investigation.

Logistic Regression vs. Random Forest

Logistic regression outperformed Random Forest on every metric at both the SES-only and full demographic levels, with AUC of 0.718 vs. 0.654. This suggests the relationships between ordinal SES predictors and binary adoption are sufficiently linear and additive that ensemble flexibility does not improve prediction and may introduce variance that hurts generalization. For survey-based classification tasks with ordinal predictors and a balanced binary outcome, logistic regression appears to be the more suitable choice.

Lessons Learned

Three methodological takeaways stand out from this project. First, survey routing exclusions can account for the majority of missing observations and represent an important generalizability limitation on the findings. Further demographic analyses should be done to assess the dropped data to see who the analytic sample represents and who it excludes. Second, applying survey weights to descriptive statistics and inferential models meaningfully changes estimates. The weighted adoption rate (47.0%) was notably lower than the unweighted rate (50.1%). Third, predicted probabilities from tree-based models in sparse feature-space regions can be unreliable. For example, the instability in the less-than-high-school heatmap row is a product of thin training data, not a real pattern.

Limitations

The binary adoption outcome collapses meaningful variation in how deeply and productively respondents use AI. A respondent who tried an AI chatbot once and never returned is coded identically to a daily user. Future work should model frequency of use as the primary outcome and include other features that were not available in the dataset, such as what the respondents are primarily using the tools for and depth of use, to understand the kind of AI engagement that actually drives the productivity premium.

Additionally, the income distribution was heavily right-skewed, with the $130k+ category nearly six times larger than some mid-range brackets, which could potentially distort estimates for those cells. The non-monotonic income gradient in the descriptive statistics, with adoption rates relatively flat between $40–80k before jumping above $100k, suggests income may have a threshold rather than linear relationship with adoption that the ordinal coding does not capture.

Future work could include the likely omitted predictors, such as occupational field, digital literacy, and social network exposure to AI, to account for much of the unexplained variance. Most significantly, the analytic sample is conditioned on chatbot awareness. The 1,364 respondents excluded by survey routing were substantially lower-SES and older than the sample used in this study, meaning the true population-level adoption gap is larger than what is reported here. Adoption rates and socioeconomic gradients in this study should be understood as describing AI-aware U.S. adults, not the full adult population

Future Directions

Future work can include replicating this analysis across multiple ATP waves to reveal whether the socioeconomic adoption gap is widening or narrowing over time as AI tools become more mainstream. The Hispanic adoption finding warrants targeted follow-up, ideally with data that include occupational field and language of AI tool use. Related published work, like Pew Research Center's February 2024 study on demographic differences in how teens use and view AI, finds that Black and Hispanic teens are more likely than White teens to use chatbots overall. Comparing these patterns among teens with the adult socioeconomic gradient identified here could reveal whether demographic predictors of AI adoption are consistent or divergent across age groups, and whether the privilege axis observed in adults is already forming in younger cohorts. Incorporating the awareness-excluded respondents through a two-stage model to first predict awareness, then predict adoption conditional on awareness would provide a more complete picture of population-level AI inequality than adoption models alone can offer.

References

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Brynjolfsson, E., Li, D., & Raymond, L. R. (2023). *Generative AI at work* (NBER Working Paper No. 31161). National Bureau of Economic Research. https://doi.org/10.3386/w31161

Lin, L. (2025, February 25). *Workers' exposure to AI.* Pew Research Center. https://www.pewresearch.org/social-trends/2025/02/25/workers-exposure-to-ai/

Lin, L. (2025, October 6). *About 1 in 5 U.S. workers now use AI in their job, up since last year.* Pew Research Center. https://www.pewresearch.org/short-reads/2025/10/06/about-1-in-5-us-workers-now-use-ai-in-their-job-up-since-last-year/

McClain, C., Anderson, M., Sidoti, O., & Bishop, W. (2026). Demographic differences in how teens use and view AI. Pew Research Center. https://www.pewresearch.org/internet/2026/02/24/demographic-differences-in-how-teens-use-and-view-ai/

Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science, 381*(6654), 187–192. https://doi.org/10.1126/science.adh2586

Pew Research Center. (2025). *American Trends Panel Wave 152* [Data set]. https://www.pewresearch.org/dataset/american-trends-panel-wave-152/

PwC. (2025a). *2025 Global Workforce Hopes and Fears Survey*. https://www.pwc.com/gx/en/issues/workforce/hopes-and-fears.html

PwC. (2025b). *AI jobs barometer 2025*. https://www.pwc.com/gx/en/services/ai/ai-jobs-barometer.html